MIL-D-16196D(SHIPS)
19 October 1960

SUPERSEDING
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30 December 1957

MILITARY SPECIFICATION

DISTILLATION UNIT, WATER, THERMOCOMPRESSION

1. SCOPE

1.1 This specification covers electrically operated distillation units of the thermocompression type for use on Naval Ships.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-R-175 - Resistance Wire.

MILITARY

MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of

MIL-B-857 - Bolts, Nuts and Studs.

MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests for

MIL-C-915 - Cable, Cord and Wire, Electrical (Shipboard Use).

MIL-E-917 - Electric Power Equipment, Basic Requirements For (Naval Shipboard Use).

MIL-D-963 - Drawings, Production, Procedure for Procurement of.

MIL-E-2036 - Enclosures for Electric and Electronic Equipment, (Naval Shipboard).

MIL-C-2174 - Controllers, Motor Starters, and Master Switches-Direct-Current Naval Shipboard Use.

MIL-C-2212 - Controllers, Motor Starters, and Master Switches - Alternating-Current, Naval Shipboard Use.

MIL-G-2860 - Glasses, Port Light, Evaporator.

MIL-P-2863 - Packing, Condenser-Tube, Flexible-Metallic (Symbol 1435).

MIL-C-3774 - Crates, Open, Wood (2,500 to 10,000 Lb.).

MIL-P-5425 - Plastic, Sheet, Acrylic, Heat Resistant.

MIL-T-15005 - Tubes, 70-30 and 90-10 Copper Nickel Alloy, Condenser and Heat Exchanger.

MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).

MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).

MIL-T-15164 - Thermometer, Self-Indicating, Liquid in Glass; Industrial.

MIL-P-15424 - Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

MIL-C-15726 - Copper-Nickel Alloy Rods, and Flat Products (Flat Wire, Strip, Sheet, Bar and Plate).

MIL-E-15809 - Expander, Tube, Condenser and Heat Exchangers.

MIL-A-15939 - Aluminum Bronze Bars, Plates, Rods, Sheets, Strips, Forgings, and Structural and Special Shaped Sections.

MIL-T-15942 - Thermometers, Distant-Reading, Indicating-Dial.

MIL-P-16298 - Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts and Associated Repair Parts.

MIL-T-16420 - Tube, 70-30 and 90-10 Copper-Nickel Alloy, Seamless and Welded.

MIL-P-16789 - Preservation, Packaging, Packing and Marking of Pumps General, and Associated Repair Parts.

MIL-M-17060 - Motors, Alternating Current, Integral Horsepower (Shipboard Use).

MIL-T-17244 - Thermometer, Self-Indicating Bimetallic, Shock-Resistant.

FSC 4620

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MILITARY (cont'd.)

MIL-M-17413 - Motors, Direct Current Integral H. P. Naval Shipboard.
MIL-A-17472 - Asbestos Sheet Compressed (Packing Material).
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-C-17728 - Control and Drive Assembly, Tube Expander, Automatic.
MIL-B-17931 - Bearings, Ball, Annular, Specially Selected For Quiet Operation (Noise Tested).
MIL-G-18997 - Gages, Pressure, Dial Indicating, Bourdon Tube.
MIL-T-19646 - Thermometers, Remote Reading, Self-Indicating Dial, Gas Actuated.
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).
MIL-P-20087 - Paint, Heat-Resisting.

STANDARDS

MILITARY

MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.

DRAWINGS

BUREAU OF SHIPS

B-214 - Root Connections for Attaching Piping.

PUBLICATIONS

BUREAU OF SHIPS

NAVSHIPS 250-000 - Bureau of Ships Technical Manual.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM - B36 - Brass Plate, Sheet, Strip, and Rolled Bar.
ASTM - B98 - Copper-Silicon Alloy Rod, Bar and Shapes.
ASTM - B111 - Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock.
ASTM - B122 - Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Alloy Plate, Sheet, Strip, and Rolled Bar.
ASTM - B139 - Phosphor Bronze Rod, Bar, and Shapes.
ASTM - B150 - Aluminum Bronze Rod, Bar, and Shapes.
ASTM - B164 - Nickel-Copper Alloy Rods and Bars.
ASTM - B171 - Copper-Alloy Condenser Tube Plates.
ASTM - D512 - Method of Test for Chloride Ion in Industrial Water and Industrial Waste Water.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York).

3. REQUIREMENTS

3.1 Qualification. - The distillation unit, including vapor compressor, furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Material. -

3.2.1 Evaporators and heat exchangers shall be of the following classes of materials as specified (see 6.1):

Class A - (see 3.19)
Class B - (see 3.20)
Class C - (see 3.21)

3.2.2 Material not definitely specified hereinafter shall be of the best quality.

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3.3 Design.- The distillation unit shall be of the vapor compression type (thermo-compression), wherein electrical power for starting and stabilizing the operation, and for operation of the vapor compressor by motor drive shall be the only source of power and heat. The units shall be furnished "knocked-down" in several component parts, all piping and wiring to be installed by the shipyard or as a self-contained "package type" unit ready for operation after installation as a unit by the shipyard, as specified (see 6.1). The units shall consist of the following:

(a) One evaporator.
(b) One or more heat exchangers.
(c) One vapor compressor with pulley and V-belts, if required, motor and controller.
(d) One set of accessories to complete the unit including pressure gages in accordance with Specification MIL-G-18997, ammeters,[1] pilot light to indicate when heaters are operating, regulating flow control and relief valves, rotameters for feed and brine overflow, and electric immersion heaters complete with automatic and manual control switches. The electric immersion heaters shall be suitable for voltages and current characteristics as specified (see 6.1).

3.3.1 Electric immersion heaters for stabilization of operation shall be installed in the distillate side of the evaporator in such manner that they will be immersed in distillate under all conditions of inclination of the ship (see 3.3.7).

3.3.1.1 Additional electric immersion heaters as required for satisfactory operation with feed water between 28°Fahrenheit (F.) and 85°F. shall be provided for installation in the feed line ahead of the heat exchanger(s). These heaters shall be contained in suitable casings of 90-10 copper nickel in accordance with Specification MIL-C-15726 or MIL-T-16420, designed for adequate circulation of feedwater around the heaters. The heaters shall be provided with electric switches (see 3.14.1), thermostatically controlled, to turn the heaters off when the temperature of the feedwater is such that additional heating over that supplied by the stabilization heaters is not required.

3.3.2 The regulating flow control valve specified in 3.3(d) shall be of the type which regulates the rate of flow of feed water to that required by the evaporator regardless of the initial pressure of the feed water, and also reduces the pressure to within safe limits for operation of the distilling unit.

[1]One ammeter for the electric heater circuit and one for the compressor motor circuit shall be furnished.

3.3.3 Rotameters specified in 3.3(d) shall be of the type in which the float is guided by flutes in the tube, rather than by a guide rod on which the float rides or which extends through a guide bushing; guide rods having been found unsuitable due to scale formation. Meters shall be provided with metal casings with safety glass windows. Float tube packing gland followers, bolts, and nuts shall be nonferrous. Inlet and outlet shall be flanged.

3.3.4 Provisions shall be made for installation of thermometers as required. Unless otherwise specified (see 6.1), thermometers shall be furnished by the shipbuilder. Thermometers shall be in accordance with Specification MIL-T-15164 for distillation units to be installed on fuel oil burning ships and in accordance with Specification MIL-T-17244 for units to be installed on nuclear propelled ships. When location of thermometers would make them difficult to read, distant reading thermometers may be substituted. Distant reading thermometers shall be in accordance with Specification MIL-T-15942 for fuel oil burning ships and in accordance with Specification MIL-T-19646 for nuclear propelled ships.

3.3.5 Distillation units for installation on submarines shall be so designed that they will not be damaged when subjected to ambient pressures of 30 pounds per square inch absolute (p.s.i.a.) maximum or 10 p.s.i.a. minimum.

3.3.6 Distillation units shall be designed for automatic operation, to require minimum attention. Units for submarines shall operate satisfactorily when the ambient pressure varies from normal atmospheric to plus or minus 6 inches of mercury.

3.3.7 Distillation units shall be designed to operate satisfactorily under the following conditions of inclination:

Surface ships:
(a) Up to 5 degrees from the normal horizontal position in the fore and aft plane (permanently inclined).
(b) Up to 15 degrees to either side (permanently inclined).
(c) With the ship rolling up to 30 degrees from the vertical to either side or pitching 10 degrees up or down from the normal horizontal plane.[1]

Submarines:	Submerged	Surfaced[2]
(a) Trim, fore and aft (permanent)	30 degrees	7 degrees
(b) List, port and starboard (permanent)	15 degrees	15 degrees
(c) Pitching	10 degrees	10 degrees

[1]The unit shall be so designed that no damage will occur if it is subjected to rolling of 45 degrees.

[2]Includes snorkel condition.

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Submarines: (cont'd.)		Submerged	Surfaced[3]
(d)	Period of complete pitch cycle	60 seconds	6 seconds
(e)	Rolling	30 degrees	30 degrees[4]
(f)	Period of complete roll cycle	12 seconds	8 seconds

[3]Includes snorkel condition.
[4]The unit shall be so designed that no damage will occur if it is subjected to rolling of 60 degrees.

Note. - Degrees are measured from the horizontal fore-and-aft plane, up or down for trim and pitch, and from the vertical to either side for list and roll. The conditions for permanent list and roll or for trim and pitch shall not be considered additive. The period is the time for one complete cycle; that is, the time for the ship to pitch or roll from one extreme to the other and return to the first extreme. The resultant angle of inclination from the following combinations shall be met:

(a) List and trim.
(b) List and pitch.
(c) Roll and trim.
(d) Roll and pitch.

3.4 Capacity and purity of distillate. -

3.4.1 Capacity. - The distillation unit shall be capable of producing an average capacity of distillate per 24 hours of operation as specified (see 6.1) over a period of 250 hours without descaling or cleaning when producing distillate from sea water of not less than 1/32 density (32 pounds of sea water containing one pound of dissolved salts), when supplied with sea water at temperatures between 28° and 85° F., with the concentration of the brine overflow maintained at not more than 4/32 density, with the saturation temperature of the compressed vapor not exceeding 230° F. and under conditions of operation specified in 3.3.6 and 3.3.7.

3.4.2 Purity of distillate. - The distillate produced shall have a salinity content not exceeding 0.0325 equivalents per million (e.p.m.) of chlorides (0.125 grains of sea salts per gallon) in units to be installed on nuclear propelled surface ships or submarines or 0.065 e.p.m. of chlorides (0.250 grains of sea salts per gallon) in units to be installed on fuel oil burning surface ships or submarines, under conditions specified in 3.3.6, 3.3.7 and 3.4.1.

3.4.3 The electric heaters shall be of sufficient wattage to enable the unit to be placed in operation within a period of 4 hours when the sea water temperature is 30° F.

3.4.4 With clean tubes in the evaporator and heat exchanger, the unit shall be capable of producing 120 percent of the capacity specified (see 6.1).

3.5 Shock resistance, sound and vibration. -

3.5.1 Complete distillation units or separately mounted components thereof weighing 4500 pounds or less shall withstand the tests specified in 4.2.3.

3.5.1.1 For distilling units weighing over 4500 pounds, the sub-bases, feet, structural members and holding down bolts shall be designed for the shock design numbers shown on figure 1. Calculations shall be submitted upon request of the bureau or agency concerned.

3.5.1.2 Nozzle flanges or piping shall not constitute the means of support of components or accessories.

3.5.1.3 Bolts designed to be stressed in shear shall be installed in holes no greater than the following sizes:

Nominal bolt diameter (inch)	Maximum diameter of hole (inch)
3/4 and smaller	Nominal bolt diameter + 1/32
Larger than 3/4	Nominal bolt diameter + 1/16

3.5.1.3.1 Units that are rigidly supported shall not be attached to two structures which can deflect relative to each other under shock loadings.

3.5.1.3.2 Shock mounts shall not be used without prior approval of the bureau or agency concerned.

3.5.1.3.3 Where braces must be employed to afford stability under vibration, the braces shall be designed to fail under a load caused by a force equal to five times the weight of the unit. This load shall be assumed to be acting at the center of gravity of the unit.

3.5.1.3.4 Where buffers must be employed to limit deflection under shock loading, the buffer shall be designed in accordance with figure 1.

3.5.2 Sound and vibration. - Distillation units to be installed on certain types of ships, such as submarines, shall not exceed the maximum allowable structure and air-borne levels for sound and vibration specified (see 6.1).

3.5.2.1 Ball bearings of rotating components of distillation units subject to sound and vibration limitations shall be in accordance with Specification MIL-B-17931, specially selected for quiet operation.

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3.6 Evaporator.-

3.6.1 The evaporator shall be composed of a shell, tube sheets, tubes, and the necessary baffles or vapor separators to insure delivery of distillate having the specified purity. The shell shall be divided into sections as follows:

(a) Vapor separator section above the upper tube sheet, which will contain the baffles or vapor separators, and will be provided with a vapor offtake opening connecting to the suction of the vapor compressor. A removable sight glass shall be provided above the upper tube sheet for observation of the water level during operation and chemical cleaning.
(b) Evaporator section with upper and lower tube sheets, with straight tubes expanded into the tube sheets at each end. Salt feed water will be brought to boiling temperature inside the tubes by heat transmission through the tube walls from fresh water vapor surrounding the tubes in the shell. Provision for inlets to the shell for the fresh water vapor shall be provided from the discharge of the vapor compressor and from the distillate boiler section. Any baffles necessary for proper direction of the vapor over the tube bundles shall be provided. A brine overflow connected to the space above the top tube sheet and set at the requisite height for maintenance of proper salt water level in the tubes shall be provided. Interconnecting of the distillate collection space in the shell with the distillate boiler section shall be provided in such manner that the minimum amount of tube surface is submerged in the distillate and the distillate boiler section is maintained filled to a level which will keep the electric heater elements submerged at all times.
(c) Salt water feed and recirculation section below the lower tube sheet.
(d) Distillate boiler section may be provided below the salt water feed and recirculation section or may be a separate component. It shall be provided with connections for insertion of the electric heating elements, distillate supply from the evaporator distillate collection space, vapor offtake connected to the evaporator section, a gage glass or sight glass, and a filling connection for start-up.

3.6.2 Sight glass assemblies using glass in accordance with Specification MIL-G-2860 shall be composed of a metal flanged frame to be bolted to a boss or flange on the evaporator shell. The frame shall contain the sight glass. The glass shall be secured with its gaskets by means of a ring screwed into the frame so as to obtain even pressure on the glass, and so that the pressure on the glass is not changed when the sight glass frame assembly is removed from the evaporator shell. Jack screws shall be provided in the flange of the frame.

3.6.2.1 Sight glass assemblies using heat resistant plastic acrylic sheet in accordance with Specification MIL-P-5425, in lieu of glass, may be composed of a single flange bolted to a boss or flange on the evaporator shell and confining the plastic sheet between resilient gaskets. Jack screws shall be provided in the flange.

3.7 Heat exchanger(s).-

3.7.1 The heat exchanger(s) may be of the following designs, as approved by the bureau or agency concerned:

(a) Double or triple tube design with straight tubes expanded or packed into header plates, with channels bolted thereto for direction of the fluid flows in the inner tube and in the annuli between the inner and outer tubes;
(b) Double or triple tube design with straight tubes interconnected as required for fluid flow with compression type heat exchanger tube fittings approved by the bureau or agency concerned.
(c) Straight tube design in which a block of high heat conductivity metal is cast around individual tubes and the tubes are expanded or packed into header plates, with channels bolted thereto for direction of fluid flow in the various tubes. In lieu of header and channel plates, tubes may be interconnected by means of compression type heat exchanger tube fittings approved by the bureau or agency concerned.
(d) Coiled tube design consisting of a flattened tube arranged in a spiral coil with a plate welded to the coil in such manner that the spaces between casing, coil, and plate form two fluid circulation channels for feed and brine, and distillate circulates in the coil.

3.8 Tubes and tube sheets for evaporator.- When tubes are expanded into tube sheets or headers, the expansion shall be accomplished by means of an automatic tube expander control in accordance with Specification MIL-C-17728, using a tube expander in accordance with Specification MIL-E-15809. The expander shall be so adjusted that the expansion is

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stopped not closer than 1/8 inch to the inner face of the tube sheet. Holes for tubes in the tube sheets or headers shall be drilled 0.001 inch larger than the nominal outside diameter of the tubes with 0.002 inch plus tolerance. Minimum ligament between holes for tubes shall be 3/16 inch. Minimum tube sheet thickness shall be 3/4 inch.

3.9 Compressor.- The compressor shall be of the 3 helical lobe positive displacement type, other positive displacement lobe type, or positive displacement water sealed type; or of the centrifugal type if for vacuum operated units. The compressor shall be all bronze, copper-nickel-alloy, or other material suitable for the purpose intended (see 3.5.2.1 for special requirements for bearings, as applicable).

3.10 Weight.- The weight of the complete unit shall be the minimum consistent with rugged design requiring minimum maintenance.

3.11 Access for installation, maintenance, or replacement.- The distilling unit shall be so designed that all parts requiring cleaning, removal, adjustment, or observation may readily be reached without removal of the unit from the foundation, except in case of major overhaul. Consideration shall be given in the design to ease of circulation and venting of gases formed when scale cleaning by means of sodium acid-sulphate, citric acid, sulfamic acid or similar solutions is resorted to as a routine cleaning measure. The tube packing used in the heat exchangers shall be resistant to the cleaning solutions specified herein.

3.12 Thermal insulation.- Unless otherwise specified the distilling unit will be insulated by the shipbuilder. (See 6.1).

3.13 Tinning.- When specified (see 6.1), all metal in contact with the distillate shall be tinned with pure tin. Drawings shall include full information as to method of tinning used and the resulting thickness.

3.14 Electrical equipment.-

3.14.1 Pressure switch, manually operated switches, controllers and master switches, and relays furnished as components of the distilling unit shall be in accordance with the applicable specifications listed in 2.1. These components shall be wired with cable conforming to Specification MIL-C-915. These components, except the pressure switch, may be mounted in a single control cabinet, class 1, drip-proof in accordance with Specification MIL-E-2036.

3.14.2 Electric heating units.-

3.14.2.1 Heating units shall consist of tubular type electric heating elements secured to cast-bronze or brass flanges by means of brazing or compression fittings. The heating elements shall consist of resistance wire conforming to composition D or E of Specification QQ-R-175, mechanically centered in magnesium oxide insulating material and enclosed in a tubular metal sheath which shall hold the wire and insulating material firmly in place. The sheath shall be nickel-copper alloy. The magnesium oxide insulating material shall be tightly packed to prevent the wire from grounding to the sheath when subjected to bending, forming, shock or vibration. At the ends of the elements, a length shall be left unheated to prevent damage to connections and wire by overheating. The ends of the elements shall be hermetically sealed by ceramic-to-metal seals. The terminals to which lead wires are attached shall be nickel-copper alloy or other metal of equal heat-resisting and corrosion-resisting qualities. Connections to the elements shall have locknuts or lockwashers. Each heating unit shall be provided with an identification plate giving connection data.

3.14.2.2 Heating units shall be designed and constructed to withstand for a period of 1 minute, a dielectric test voltage of twice rated voltage plus 1000 volts between the heating elements and ground, applied continuously for a period of not less than 60 seconds. The frequency shall not be less than 60 cycles and the voltage wave shall approximate a sine wave.

3.14.2.3 The insulation resistance of each heating unit, when at normal operating temperatures shall be not less than 2.0 megohms. The insulation resistance at normal room temperature shall be not less than 4.0 megohms.

3.14.3 Motors and controller.- Motors and controllers shall be furnished in accordance with Specifications MIL-M-17060 or MIL-M-17413 and MIL-C-2174 or MIL-C-2212, respectively (see 3.5.2.1 for special requirements for bearings, as applicable).

3.14.4 Extra precautions shall be taken in the installation of electrical apparatus to insure that it is mounted away from possible sources of water, oil, or moisture damage. Adequate clearance shall be provided in the immediate vicinity of electrical equipment to permit and facilitate maintenance work. All electrical equipment shall comply with the requirements of Specification MIL-E-917.

3.15 Threaded parts.- Threaded parts shall conform to Handbook H28. Designs shall permit the use of standard wrenches throughout. Cap screws shall not be used.

3.16 Welding.- Welding shall be in accordance with Standard MIL-STD-278.

3.17 Root connections.- Root connections shall be in accordance with Drawing B-214 or nozzles may be fabricated of short lengths of pipe provided with a slip-on flange welded to one end or pads may be of bar stock or plate of requisite thickness for studs. Nozzles shall be of minimum length possible for through bolts.

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3.18 Flanges.- Flanges for pipe connections shall be 100 pounds per square inch in accordance with Specification MIL-F-20042, except those flanges in submarine installations which are subjected to submergence pressure. Such flanges shall be as specified (see 6.1).

3.19 Class A materials.- Class A materials for evaporators and heat exchangers shall be in accordance with table I.

Table I - Materials for evaporators and heat exchangers

Application	Material	Applicable document
Evaporators:		
Shell, baffles and tube sheets	Copper-nickel alloy, alloy No. 5[1]	ASTM B-122
Tubes	Copper-nickel alloy, composition 70-30	MIL-T-15005
Tubing	Copper-nickel alloy, composition 70-30	MIL-T-16420
Bolts and nuts	Nickel-copper alloy, class A	ASTM B164
	or bronze, aluminum; wrought, alloy 3, stress relieved	ASTM B150
	or phosphor bronze, alloy A or D	ASTM B139
	or copper-silicon alloy	ASTM B98
Bolts, form	----------	MIL-B-857
Gaskets	Synthetic rubber, cloth insertion, commercial	----------
	or compressed asbestos	MIL-A-17472
Heat exchangers:		
Casing (if not in contact with salt water)	Naval wrought brass	ASTM B171
	or brass	ASTM B36
Casing (if in contact with salt water)	Copper-nickel alloy, alloy No. 5[1]	ASTM B122
Tubes in contact with salt water feed or brine overboard	Copper-nickel alloy, composition 70-30	MIL-T-15005
Tubes in contact with distillate only	Copper	ASTM B111
Bolts and nuts	Nickel-copper alloy, class A	ASTM B164
	or bronze, aluminum wrought, alloy 3, stress relieved	ASTM B150
	or phosphor bronze, alloy A or D	ASTM B139
	or copper-silicon alloy	ASTM B98
Bolts, form	----------	MIL-B-857
Gaskets	Synthetic rubber, cloth insertion, commercial	----------
	or compressed asbestos	MIL-A-17472
Tube packing	Metallic, grade A	MIL-P-2863
Tube sheets and water channel plate, when in contact with salt water.	Copper-nickel alloy, alloy No. 5[1]	ASTM B122
Tube sheets and water channel plate, when in contact with distillate only	Naval wrought brass	ASTM B171

[1]Iron content 0.4-0.7 percent (0.5 percent nominal).

3.20 Class B materials.- Class B materials for the evaporator and heat exchangers shall be the same as for class A except copper-nickel alloy shall be composition 90-10 and Specification MIL-C-15726 shall be used instead of ASTM B-122, where specified.

3.21 Class C materials.- Class C materials for evaporators and heat exchangers shall be in accordance with table II.

3.21.1 All components fabricated by welding when made of bronze, aluminum, wrought, shall be adequately heat treated after fabrication to avoid stress corrosion cracking.

3.22 Drawings.- Drawings shall be furnished in accordance with the requirements of Specification MIL-D-963.

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Table II - Materials for evaporator and heat exchanger.

Application	Material	Applicable document
Evaporators:		
Shell, baffles and tube sheets	Bronze, aluminum, wrought, composition 5	MIL-A-15939
Tubes	Copper-nickel alloy, composition 90-10	MIL-T-15005
Tubing	Copper-nickel alloy, composition 90-10	MIL-T-16420
Bolts and nuts	Bronze, aluminum wrought, stress relieved, alloy 3	ASTM B150
	or phospher bronze, alloy A or D	ASTM B139
	or copper-silicon alloy	ASTM B98
Bolts, form	----------	MIL-B-857
Gaskets	Synthetic rubber, cloth inserted, commercial	----------
Gaskets	Compressed asbestos	MIL-A-17472
Heat exchanger:		
Casing, (if not in contact with salt water)	Naval rolled brass	ASTM B171
	or brass	ASTM B36
Casing, (if in contact with salt water)	Bronze aluminum, wrought, composition 5	MIL-B-15939
Tubes in contact with salt water feed or brine overboard	Copper-nickel alloy, composition 90-10	MIL-T-15005
Tubes in contact with distillate only	Copper	ASTM B111
Bolts and nuts	Bronze, aluminum, wrought, stress relieved, alloy 3	ASTM B150
	or phosphor bronze, alloy A or D	ASTM B139
	or copper-silicon alloy	ASTM B98
Bolts, form	----------	MIL-B-857
Gaskets	Synthetic rubber, cloth inserted, commercial	----------
Gaskets	Compressed asbestos	MIL-A-17472
Tube packing	Metallic, grade A	MIL-P-2863
Tube sheets and water channel plate	Bronze aluminum, wrought, alloy 3	ASTM B150

3.22.1 Types of drawings.- In lieu of the drawing types listed in Standard MIL-STD-7, the following types are required:

3.22.1.1 Outline drawing.- An external arrangement drawing shall show all necessary external views of the unit, shall include all external dimensions required for reproduction on ship's machinery arrangement drawings for guidance of the shipyard in designing the foundation structure for the unit, for installation thereof, and for connection of the unit to the external piping. The drawing shall show the space required for removal and replacement of tubes in the heat exchanger components, removal of waterboxes, covers of access openings, and electric immersion heaters, location of lifting lugs or eyebolts provided for handling of the unit or its components and size of openings therein provided for lifting.

3.22.1.2 Assembly Drawing.- A drawing showing complete longitudinal and transverse cross-sectional views of the unit shall also show the relationship of all parts, arrangement of tubes in heat exchanger components, method of tube end expansion, baffles, vapor separators, and welding of adjacent parts. Liberal use of enlarged views or sections shall be made. If necessary, subassembly drawings conforming to the above may be furnished for individual components of the unit. The drawing or drawings shall be such that a thorough understanding of the design and construction of the apparatus may be obtained without reference to related detail drawings.

3.22.1.2.1 The assembly drawing shall contain a list of materials showing names of parts with identifying numbers and materials of all parts (see 3.2). The identifying numbers shall also be shown adjacent to the part depicted in the various views, with arrows pointing to the parts.

3.22.1.3 Detail drawings showing all major parts such as shells, waterboxes, tube sheets, tube supports, baffles, vapor separators, and the like. The drawings shall be completely dimensioned, with

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finishes and welding symbols indicated, as required for manufacture.

3.22.1.4 A diagrammatic piping arrangement drawing showing complete piping required for the operation of the apparatus, with piping furnished with the apparatus and that to be furnished by the shipbuilder clearly indicated. Location of gages, thermometers, valves, acid cleaning connections, and salinity indicators shall be indicated.

3.22.1.5 A wiring and connections drawing or drawings shall be provided for such electrical components as are furnished as part of the unit.

3.22.1.6 It is the intent that the drawings furnished be sufficient to enable:

(a) Determination of compliance with this specification.
(b) Evaluation of the suitability of the design for the intended use.
(c) Evaluation of performance and maintenance capability.
(d) Installation by shipyard without supplier's assistance.
(e) Ship and shore activities to repair and maintain the apparatus.

3.22.1.7 Provided the assembly drawing(s) required by 3.22.1.2 include sufficient information to enable the drawing approval agency to act in accordance with 3.22.1.6 (a), (b) and (c), submission of detail drawings for approval is not required. They shall, however, be provided as part of the finished drawings to enable compliance with 3.22.1.6(e).

3.22.2 Ship's drawing.- One drawing shall be a ship's drawing. It shall be titled "DRAWING LIST, PERFORMANCE DATA, AND ONBOARD REPAIR PARTS LIST FOR _____".

3.22.2.1 The "drawing list" tabulation shall include the following columns:

(a) Drawing title.
(b) Manufacturer's drawing number.
(c) Bureau or agency drawing number.
(d) Revision symbol.

This list shall include all drawings which comprise the given design. The revision symbol column shall be kept up to date to the time of manufacture so that it will finally indicate the latest revision of each drawing applicable to the equipment as built.

3.22.2.2 The "onboard repair parts list" shall show the onboard repair parts and tools furnished with the unit (see 3.22). This tabulation shall include the following columns:

(a) Piece number.
(b) Name of piece.
(c) Quantity.
(d) Manufacturer's drawing number.
(e) Manufacturer's service part number[1].
(f) Weight. (Total of this column shall indicate unboxed weight of parts.)

[1]The manufacturer's service part number is defined as a constant number assigned by the manufacturer, which is the infallible identification of a part wherever used. When it is the practice to assign part numbers to all pieces, these shall be recorded on the proper drawings. Where a complete part numbering system is not in use, the manufacturer shall assign service part numbers to each distilling unit and to other parts or subassembly which in his judgment may require replacement during the life of the unit. Such numbers shall then be recorded on the proper drawings.

3.23 Repair parts.- Lists of stock repair parts and on-board repair parts shall be developed in accordance with Specification MIL-P-15137.

3.24 Onboard repair parts.- Onboard repair parts and special tools shall be furnished as follows:

3.24.1 Onboard repair parts.-

(a) 1 set of V-belts.
(b) 100 percent gaskets.
(c) 1 set electric heaters.
(d) 100 percent gage, sight and flow meter glasses.
(e) 1 gage of each size and type.
(f) 1 thermometer of each size and type (to be furnished by shipbuilder unless otherwise specified (see 6.1).
(g) 10 percent of special tubes for heat exchanger (if included in the design), including tube packing.
(h) 1 ammeter of each size and type.
(i) 1 set of ball bearings for vapor compressor.
(j) 1 set of shims for vapor compressor.
(k) 1 set of packing rings for vapor compressor.
(l) Repair parts for motors, controllers, switches, and relief valves shall be furnished in accordance with the requirements of the applicable specifications for such components.
(m) 10 percent of compression type heat exchanger tube fittings (if used).
(n) Repair parts for feedwater pressure and flow regulating valve.

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3.24.2 Special tools.- One complete set of special tools which are required for proper service maintenance shall be supplied to each vessel, including vertical basket removal tool, if required. Special tools are defined as those tools not listed in the Navy Stock List of General Stores (Copies of this stock list may be consulted in the office of the Government inspector.)

3.25 Technical manuals.- Technical manuals, in accordance with type II of Specification MIL-M-15071, shall be furnished.

3.25.1 During original preparation of the manual, the text shall be so written and arranged that the maximum possible number of individual pages thereof can be used in subsequent manuals.

3.25.2 The operating instructions should be written in connection with the manufacturer's diagrammatic drawing, identifying valves and pipe lines by name in the text, so that such instructions can be used in subsequent manuals. If the shipbuilder's diagrammatic is included in the manual, it should be supplementary and the text shall not be required to reflect the designations of valves and pipe lines indicated on such diagrammatic. Maximum density of brine overflow (thirty-seconds) or ratio of distillate to brine overboard shall be indicated.

3.25.3 The specified section on installation may be omitted unless there is something special in regard to installation which should be called to the attention of the activity concerned.

3.26 Identification plates.- Identification plates of sheet or cast brass or bronze shall be provided. Identification plate data shall include the following:

(a) Manufacturer's name.
(b) Government contract number.
(c) Government stock number (allow 17 spaces).
(d) Date of manufacture.
(e) Blank space for Government inspector's stamp.
(f) Blank space for unit number (2 spaces to be stamped by shipyard).

3.27 Painting.- Painting of external non-ferrous castings, piping, or other parts is not required. All external unmachined surfaces of ungalvanized ferrous parts shall be thoroughly cleaned and coated with two coats of heat resisting paint in accordance with Specification MIL-P-20087 or it's commercial equivalent.

3.28 When any specification which forms a part of this specification requires that the product be subjected to and pass qualification tests, only products which have been approved for listing on the applicable Qualified Products List on the date of invitation for bids or date of initiation of purchase action shall be utilized. In the event no Qualified Products List has been issued, the contractor shall request instructions as to what testing will be required to determine whether the product meets the requirements of the specification.

3.29 Workmanship.- Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests[1].- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. These tests shall consist of the tests specified in 4.2.1, 4.2.2, 4.2.3 and 4.3.

4.2.1 The unit offered for test shall be evaluated as to suitability of design and amount of power required per pound of distillate produced.

4.2.2 The unit shall be operated under the conditions specified in 3.4.1 to determine compliance with the specified conditions. Units which are small enough to be mounted on the rocking platform available in the laboratory shall be mounted thereon, and shall comply with the above when subjected to roll and pitch conditions specified in 3.3.7. Larger units shall comply with the conditions specified herein when subjected to a stationary test only.

4.2.2.1 Salinity content of the distillate shall be determined as specified in 4.3.5.3.1 during the qualification test.

4.2.3 Shock tests of complete distillation units or separately mounted components thereof weighing 4500 pounds or less, shall be performed in accordance with Specification MIL-S-901 with the following modification:

(a) Under the test procedure for medium weight equipment, the total number of blows shall be increased to nine. The nine blows shall consist of three groups

[1]Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

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of three blows each. For each group the height of hammer drop and the initial up travel of the anvil table shall be as shown in table I of Specification MIL-S-901.

(1) If feasible within weight limitations of the medium machine, one blow of each group shall be applied parallel to each of three principal axes of the equipment.

(2) If adaptors to permit the above would result in exceeding machine limitations, each group shall comprise the following:

a. The first blow shall be applied with the unit on the standard (horizontal) adaptor.

b. The second blow shall be applied with the equipment rotated 45 degrees around its principal axis (using a 45 degree adaptor).

c. The third blow shall be applied with the equipment rotated to incline the principal axis 45° to the horizontal (using a 45° adaptor).

4.2.3.1 Waiver of shock testing will be considered for units if they are of similar weight to units which have successfully passed the tests specified in 4.2.3.

4.3 Acceptance tests.-

4.3.1 Hydrostatic pressure.- All parts of the distilling unit shall be subjected to a hydrostatic pressure test as follows:

Name of part	Test pressure, pounds per square inch gage
Evaporator and vapor compressor	10
Heat exchanger	50
Other parts not specified	1-1/2 times working pressure.

4.3.2 Each electric heating unit shall be tested for conformance with 3.14.2.2 and 3.14.2.3.

4.3.3 Performance.- Each distilling unit shall be operated in the manufacturer's shop or test facility to demonstrate that the clean tube capacity is 120 percent of the specified capacity and that all component parts are in proper operating condition.

4.3.4 Sound and vibration tests.- Sound and vibration tests shall be as specified (see 6.1).

4.3.5 Performance tests after installation.- After installation, at least three capacity tests shall be made on each distillation unit as follows:

4.3.5.1 The first capacity test specified in 4.3.5 shall be made in the presence of the inspector prior to the official trials of the ship, using feed water as available at the ship's berth. The capacity shall be as specified in 4.3.3. Should it be considered that the water available is such as to foul the tubes of the unit, this test may be waived by the Government inspector.

4.3.5.2 The second capacity test specified in 4.3.5 shall be made during the official preliminary trials of the ship, the sea water used for feed being not less than 1/32 density. The capacity shall be as specified in 4.3.3.

4.3.5.3 The third capacity test specified in 4.3.5 shall be made before final acceptance of the vessel. Starting with clean tubes (chemically cleaned, if necessary, in accordance with article 58-315 of Bureau of Ships Technical Manual, NAVSHIPS 250-000), the unit shall be operated over a period of 250 hours, which shall be as nearly continuous as practicable, with feed water of not less than 1/32 density, density of brine overboard not exceeding the maximum specified by the manufacturer in the technical manual for the unit, and with feed water between 28° F. and 85° F., as available, it shall be demonstrated that the average capacity over the above specified period of operation is not less than the specified rated capacity (see 3.4.1), that the purity of the distillate is as specified in 3.4.2, and that the compressor motor is not overloaded during the specified period of operation.

4.3.5.3.1 In case of any question as to the accuracy of the determination of the salinity content of the distillate by the methods available aboard the ship, a sample of the distillate shall be submitted to a qualified laboratory for determination of the salinity content by referee method A - mercurimetric titration in accordance with ASTM Publication D512.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Basic equipment and accessories.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion and deterioration during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

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5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination. In addition, distillation units meeting the requirements of 3.5.2 shall have the exterior shipping containers marked as follows:

5.1.1.3.1 All containers shall be stenciled on two sides and both ends with minimum 1-1/2-inch letters:

CRITICAL, HIGH VALUE, CLOSE TOLERANCE
OPERATING EQUIPMENT
HANDLE WITH EXTREME CARE
AVOID UNNECESSARY MOVEMENTS, JARRING, ETC.

Arrows and "THIS SIDE UP" shall be marked on all containers.

5.1.1.3.1.2 Unpacking instructions shall be provided and shall be sealed in waterproof envelopes which shall be tacked, stapled or otherwise securely attached to the outside of each container in the most protected location. The words "UNPACKING INSTRUCTIONS" shall be stenciled on each container to indicate location of such instructions.

5.1.2 Onboard repair parts.- Onboard repair parts shall be prepared for the level of shipment as specified (see 6.1) in accordance with Specifications MIL-R-196, MIL-P-16789, MIL-P-16298, MIL-P-17555 or MIL-P-15424, as applicable.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking shall be as specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1):

5.2.1.1 Preservation and packaging.-

5.2.1.1.1 Level A.- Cleaning, preservation, and packaging procedures and methods specified herein shall be in accordance with Specification MIL-P-116.

5.2.1.1.1.1 Distilling units, vapor compression.- All unpainted ferrous metal parts shall be cleaned and preserved as specified in 5.2.1.1.2 and 5.2.1.1.3. Distilling plants of nonferrous construction require no internal preservation but all connections and openings should be properly sealed as specified in 5.2.1.1.1.1.1 to prevent the entry of foreign matter.

5.2.1.1.1.1.1 Where practicable, all openings such as inlet and outlet water connections, pipe taps, and pipe openings shall be covered with a blank flange or suitable plug, using grade A paper of Specification MIL-B-121 between the cover and metal surfaces.

5.2.1.1.2 Cleaning.-

5.2.1.1.2.1 Metal surfaces.- All unpainted or machined metal surfaces subject to corrosion shall be cleaned immediately prior to application of the preservative in such a manner that no corrosion will occur on the cleaned surface during the interval from cleaning to the application of the preservative. No petroleum base cleaning solvent shall be used on surfaces or parts which come in contact with oxygen.

5.2.1.1.2.2 Large assemblies or parts.- Unpainted metal surfaces or large assemblies subject to corrosion shall be cleaned by alternatively brushing or spraying with solvent in accordance with Specification P-S-661, and wiped dry with a clean cloth.

5.2.1.1.2.3 Disassembled parts, subassemblies, or onboard repair parts.- Disassembled parts, subassemblies, or onboard repair parts shall be cleaned by any petroleum solvent method specified in Specification MIL-P-116, which will accomplish thorough cleaning without damage to the item.

5.2.1.1.3 Preservation of ferrous metal parts subject to corrosion.-

5.2.1.1.3.1 Internal ferrous metal parts which operate in or are lubricated with oil.- With the exception of ball or roller bearings, all internal ferrous metal surfaces including main journals, or subassemblies, which operate in or are lubricated with oil or petroleum products, shall be brushed, flushed, or sprayed with a type P-2 preservative. Ball or roller bearings shall be treated with a preservative conforming to type P-6. No petroleum base preservative shall be used on surfaces or parts which come in contact with oxygen.

5.2.1.1.3.2 External ferrous metal surfaces.- All external unpainted metal surfaces of assembled components shall be brushed or sprayed with preservative conforming to grade 4 of Specification MIL-C-16173. Care must be exercised to prevent the preservative from coming in contact with leather, rubber, electrical insulation, or painted surfaces.

5.2.1.2 Repair parts, tools and accessories.- Repair parts and tools shall be prepared for the level of shipment as specified (see 6.1) in accordance with Specifications MIL-R-196, MIL-P-16789,

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MIL-P-16298, MIL-E-17555 or MIL-P-15424, as applicable. Accessories which require separate packing for proper preservation and security in shipment shall be packaged and packed as follows, and shall be included with the complete unit inside its shipping container:

(a) Mechanical parts and tools shall be preserved, packaged and packed in according with Specification MIL-R-196 and MIL-P-15424 and in accordance with Specification MIL-P-116 when not specifically covered therein.
(b) Electrical parts shall be preserved, packaged and packed in accordance with Specifications MIL-P-16298 and MIL-E-17555, as applicable.

5.2.1.3 Packing.-

5.2.1.3.1 Level A and B.- Distilling units cleaned and preserved as specified herein shall be packed in wood-cleated plywood or nailed wood boxes, sheathed or unsheathed crates conforming to Specification PPP-B-601, PPP-B-621, MIL-C-104, MIL-C-3774 or MIL-C-132, respectively.

5.2.1.3.1.1 Difficult loads.- Difficult loads, as defined in Specification JAN-P-100 shall be packed as follows:

5.2.1.3.1.2 Under 500 pounds.- Difficult loads under 500 pounds shall be packed in modified style 2 nailed wood boxes conforming to Specification PPP-B-621. This shall be accomplished by interior battens and exterior skids. When necessary, side panels shall be supported by interior diagonal members and the top reinforced by cross battens properly supported.

5.2.1.3.1.3 Over 500 pounds.- Difficult loads over 500 pounds shall be packed in modified style 2 nailed wood boxes conforming to Specification PPP-B-621 with skid type bases or in sheathed crates conforming to Specification MIL-C-104.

5.2.1.3.1.4 Liners.- Unless otherwise specified in the contract or order, shipping containers, except sheathed crates, shall contain a sealed waterproof bag or its equivalent (shrouded 2/3 of the way down and secured when a sealed waterproof bag liner is impracticable) made of material conforming to Specification UU-P-271. The seams and closures shall have strength and water resistance equal to that of the body material. Shrouds shall be so applied as to prevent the formation of water pockets and permit the free circulation of air. All sharp projects which would tend to tear the shroud shall be padded or cushioned. The top and sides of the unit being shrouded shall be sufficiently covered to prevent the entrance of water, but shall not be sealed to such an extent that air is not allowed to circulate freely.

5.2.1.3.1.5 Strapping.- Shipping containers shall be strapped in conformance with the appendix of the applicable container specification.

5.2.1.4 Technical manuals.- Technical manuals shall be prepared for the level of shipment as specified (see 6.1) in accordance with Specification MIL-M-15071.

5.2.1.6 Marking.- In addition to any special marking required (see 6.1), interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

5.2.1.6.1 Distillation units meeting the requirements of 3.5.2 shall have the exterior shipping containers marked as follows:

5.2.1.6.1.1 All containers shall be stenciled on two sides and both ends with minimum 1-1/2 inch letters:

CRITICAL, HIGH VALUE, CLOSE TOLERANCE OPERATING EQUIPMENT
HANDLE WITH EXTREME CARE
AVOID UNNECESSARY MOVEMENTS, JARRING, ETC.

Arrows, as indicated in Standard MIL-STD-129 and "THIS SIDE UP" shall be marked on all containers.

5.2.1.6.1.2 Unpacking instructions shall be provided and shall be sealed in waterproof envelopes which shall be tacked, stapled or otherwise securely attached to the outside of each container in the most protected location. The words "UNPACKING INSTRUCTIONS" shall be stenciled on each container to indicate location of such instructions.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class of material required for evaporators and heat exchangers (see 3.2.1).
(c) Whether the units shall be furnished "knocked-down" or package type (see 3.3).
(d) Voltage and current characteristics, type of insulation, degree of enclosure, service rating and other ordering data required by the applicable electrical specifications (see 3.3(d) and 3.14).
(e) Whether thermometers are to be furnished with the distillation units (see 3.3.4 and 3.24.1).
(f) Rated capacity required (see 3.4.1 and 3.4.4).
(g) Number of units per ship and number of ships.

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(h) Specifications for limitation of sound or vibration, or both, if required (see 3.5.2 and 4.3.4).
(i) Whether thermal insulation is to be provided with the unit (see 3.12).
(j) Whether surfaces in contact with distillate (required only on Diesel driven submarines) are to be tinned (see 3.13).
(k) Specifications for flanges, if other than for 100 p.s.i. (see 3.18).
(l) Level of shipment for onboard repair parts (see 5.1.2).
(m) Preservation, packaging, packing or marking required other than those required by 5.1 (see 5.2).
(n) Level of shipment for technical manuals (see 5.2.1.4).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the times set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 16196, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D.C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pa.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy - Bureau of Ships
(Project 4620-N011Sh)

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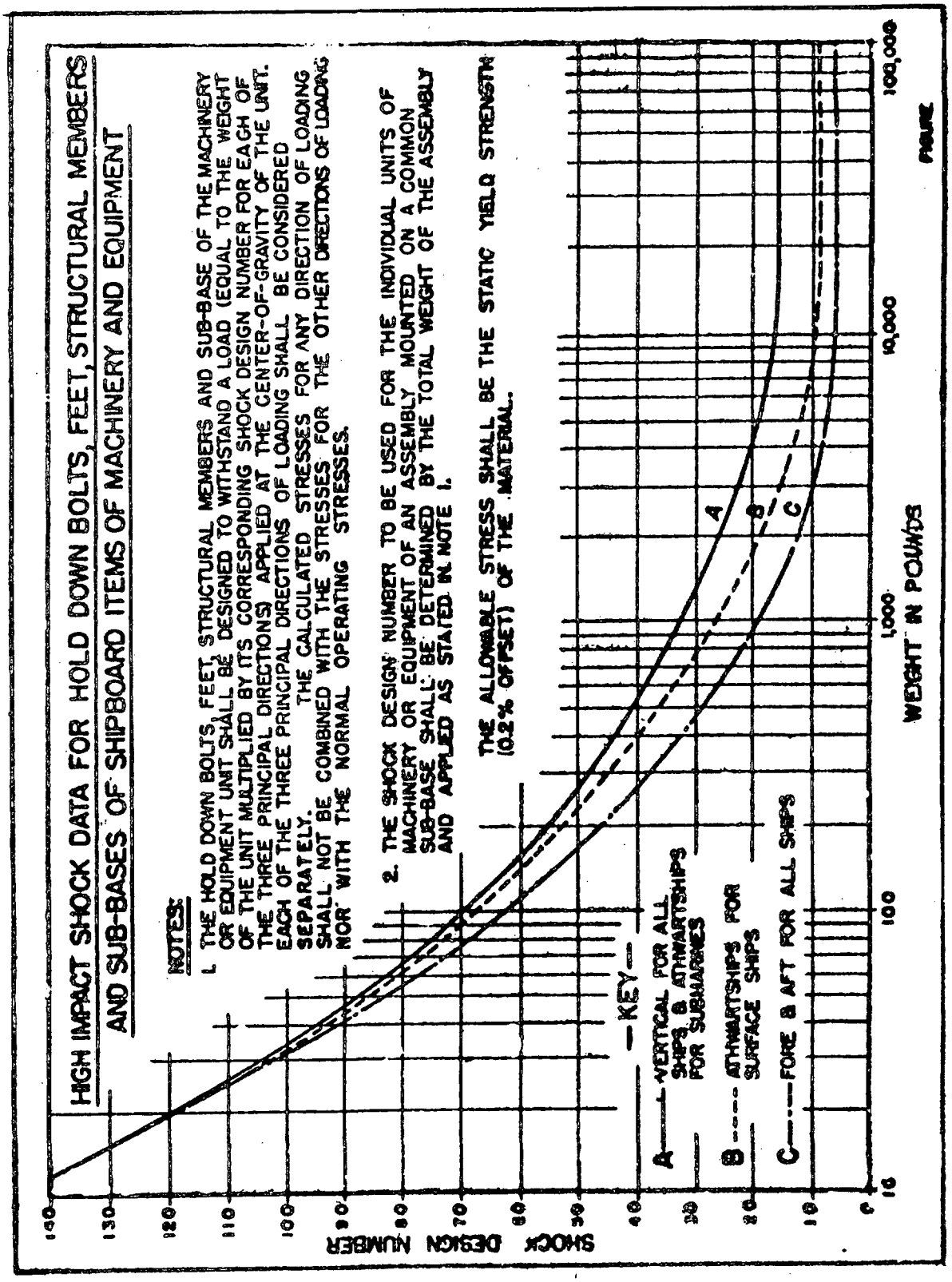


Figure 1 - H.I. Shock design data

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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